

Water Resources Management

Katie Markovich, EPS 522/ ENVS 423L (Fall 2025)

Housekeeping

- No quiz today, HW 8 will be a free 100
- HW 9 will be a peer evaluation of at least 3 of the EPS 522 term projects. You can get an extra 20% for every additional evaluation you do.
- Next Tuesday will be optional for EPS 522 students who want to work on their projects.
- Review for exam Thursday, ^{Dec}~~Nov~~ 4
- Final exam Tuesday, ⁹~~8~~ Dec. Details to come.

EPS 522 Project Presentations

- 12-minute presentation slot with 3 minutes for questions
- 1-2 minutes per slide, roughly 8 slides
 - 1 Control volume motivation & background
 - 2 Conceptual water balance
 - 3 Data sources & analysis (what did you do)
 - 4 Results (quantified components, uncertainty/variability, limitations)
 - 5 Future work
- Keep text on slides at a minimum or iterate through
- Explain every plot
- Practice for timing

Definition

Water-resource management (WR Mgmt) is the process by which governments, businesses, and/or individuals reach and implement decisions that are intended to affect future availability and/or quality of water for beneficial uses or the risk of water-related hazards that threaten beneficial activities.

Beneficial use is a legal term and means any use of water that is perceived to increase individual, business, or societal well-being.

Water Resources Management

While hydrologic and engineering sciences are a critical part of WR mgmt, they are not the ultimate determinants of **decisions**.

Hydrologic sciences

Climate
Physics
Chem
Geosciences
Biology
Math/Statistics

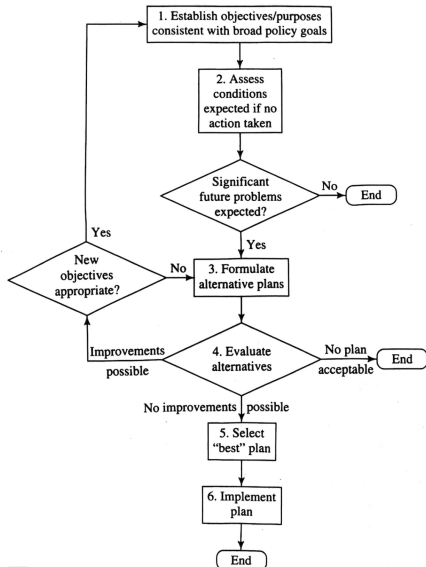
Engineering Sciences

Fluid Mechanics
Hydraulics
Meteorology*

Economic and social sciences

Law
Policy / Politics / Planning
Economics
Sociology

WR Mgmt Process & Categories



1 Water Supply & Demand

2 Water Quality

3 Floods

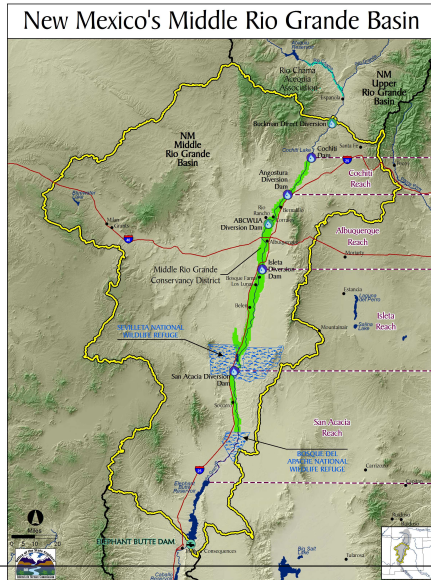
4 Low Streamflows and Droughts

Hydrologic analysis mostly plays into assessing no action conditions and evaluating alternatives.

Middle Rio Grande Basin Case Study

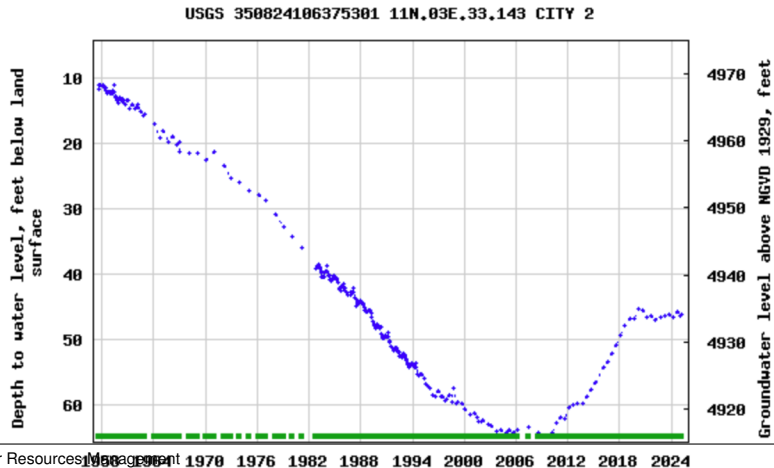
Most densely populated region of New Mexico (50% of state population). Diverse uses of surface and groundwater include:

- Irrigation
- Municipal/Domestic
- Industrial
- Ecosystem (MRGB has two species of particular concern: silvery minnow and southwestern willow flycatcher)
- Ceremonial



ABQ Water Supply Crisis

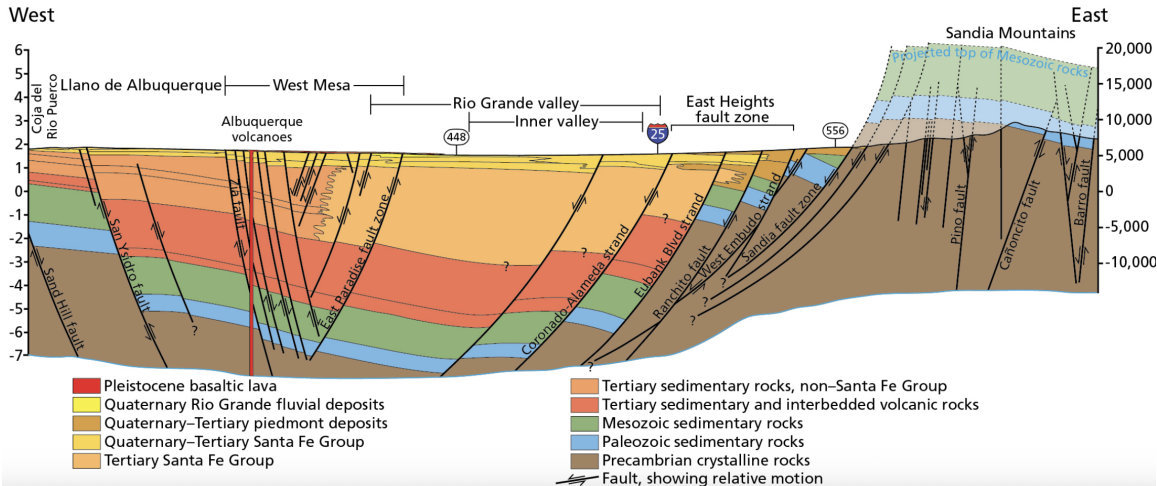
Albuquerque was entirely dependent on GW for much of its history, and until the early 90's, the prevailing thought was that we had an enormous aquifer with an equivalent storage to Lake Superior.



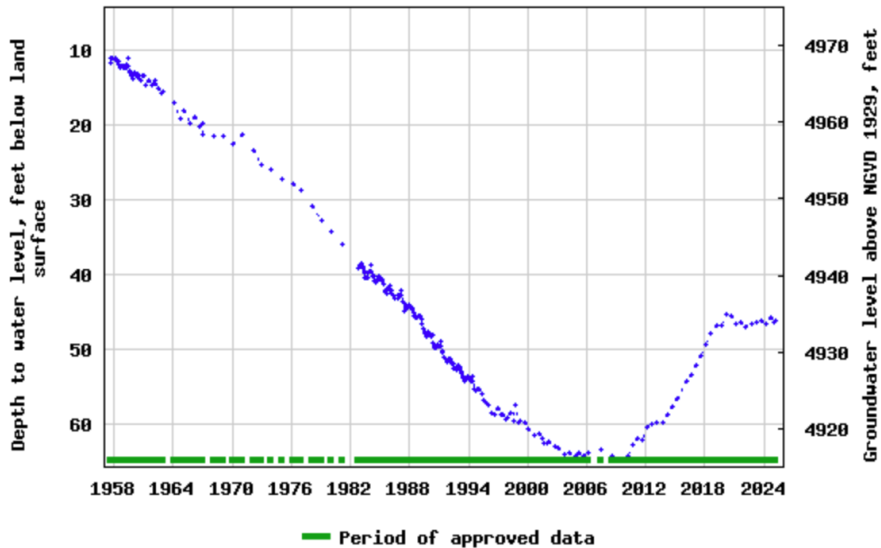
WR Mgmt Discussion

- What is the no-action condition?
- What kind of data/hydrologic analysis would inform this?
- What are the alternatives?
- What are the tradeoffs?

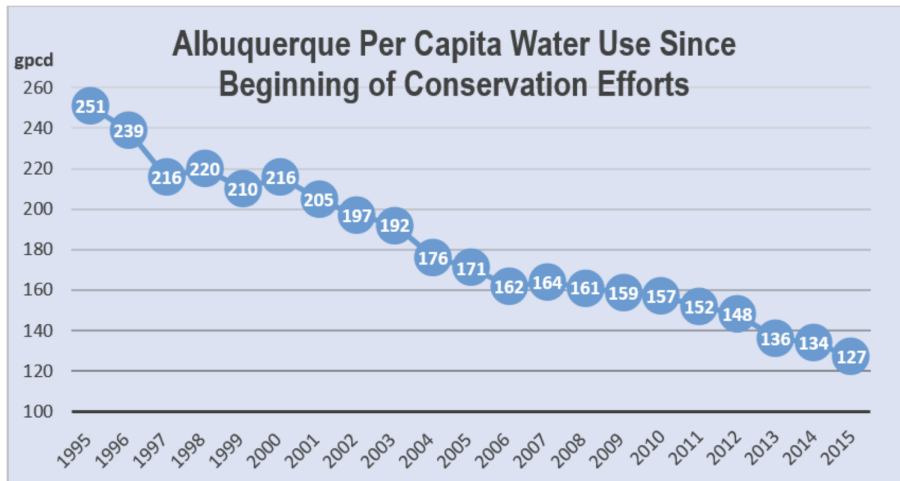
Hydrogeologic Analysis



USGS 350824106375301 11N.03E.33.143 CITY 2



Demand-Side Measures



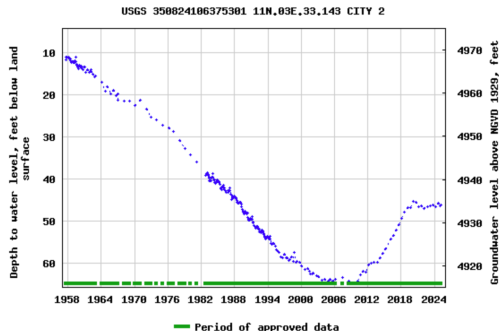
Supply-Side Measures



San Juan Chama Drinking Water Project



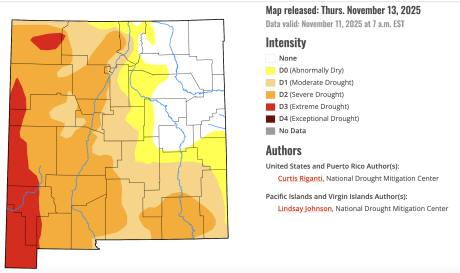
Due to drought (and climate change), ABQ and Santa Fe have not received full San-Juan Chama allotment since before 2014. Less than 1/3 of total annual water supply to ABQ.



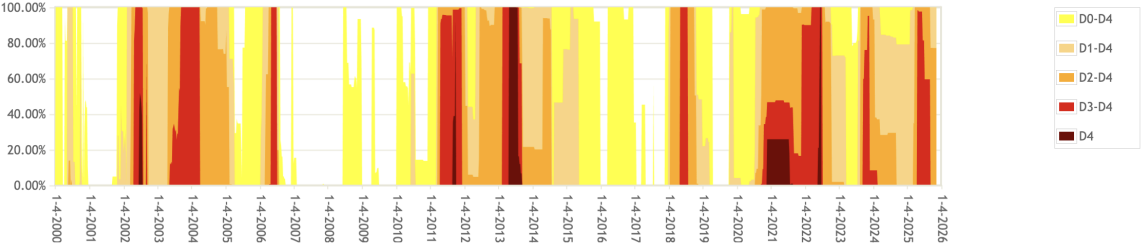
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Drought Index



Percent Area in U.S. Drought Monitor Categories



ABCWUA Drought Index

GPCD: gallons per capita consumed per day

DSCI: drought severity coverage index, add percentages of D0 through D4, values from 0 (0% area in drought) to 500 (100% area in D4)

Table 4. Drought Stages Based Criteria Chart

Groundwater Production / GPCD	DSCI ≥ 300	Less than 120% of the Annual GW Production Goal	Between 120% and 130% of GW Production Goal	Between 130% and 140% of GW Production Goal	More than 140% of the GW Production Goal
0 to < 2 GPCD over the goal	Stage 0	Stage 0	Stage 0	Stage 0	Stage 1
2-4 GPCD over the goal	Stage 0	Stage 0	Stage 1	Stage 1	Stage 2
4-6 GPCD over the goal	Stage 0	Stage 0	Stage 1	Stage 2	Stage 3
> 6 GPCD over the goal	Stage 0	Stage 1	Stage 2	Stage 3	Stage 3

ABCWUA Drought Demand Measures

STAGE 0 - DROUGHT ADVISORY

- Increase public education: Offer Drought Smart \$20 rebate classes
- Develop conservation guidelines using up-to-date consumption data
- Develop Public Drought Campaign
- ▲ Monitor regional drought conditions, and track annual supply and demand triggers

STAGE 1 - DROUGHT WATCH

- ★ Enact Water Waste Compliance Drought Measure: Double Fees for wasting water - Malfunctions should be fixed within 48 hours (about 2 days)
- Implement the Drought Awareness Campaign
- Coordination of the Drought Agency Group -1. synchronize public education and public support for vulnerable populations and 2. work with public agency partners-initiate water use audits and optimize water efficiency plans
- Send targeted information to customers exceeding conservation guidelines; water audits offered
- Expand leak alert program using AMI reports

STAGE 2 - DROUGHT WARNING

- Increased Drought Awareness Campaign
- ★ Customers exceeding conservation guidelines receive warnings of double surcharges. An audit is offered to avoid the penalty.
- ★ Enact more Water Waste Compliance measures
 - Implement low income leak/water waste assistance program
 - Double Fees continue for water waste
 - Malfunctions must be fixed in 48 hours (about 2 days) to avoid triple fees
 - Activate AMI water waste reporting system
 - No Variances granted for the establishment of new sod or turfgrass seeds
- Implement Drought Agency Group Mitigation Strategies and Programs: Identify financial assistance to implement results of audits with public agency partners

STAGE 3 - DROUGHT EMERGENCY

- Expand Drought Awareness Campaign
- ★ Customers exceeding conservation guidelines receive triple surcharges.
- ★ Irrigation limited to one day per week
- Implement 20% rebate
- ★ Triple fees for all water waste
- ★ Spray irrigation prohibited on non-functional turfgrass
- ★ Restrict non-essential water uses

Water Quality

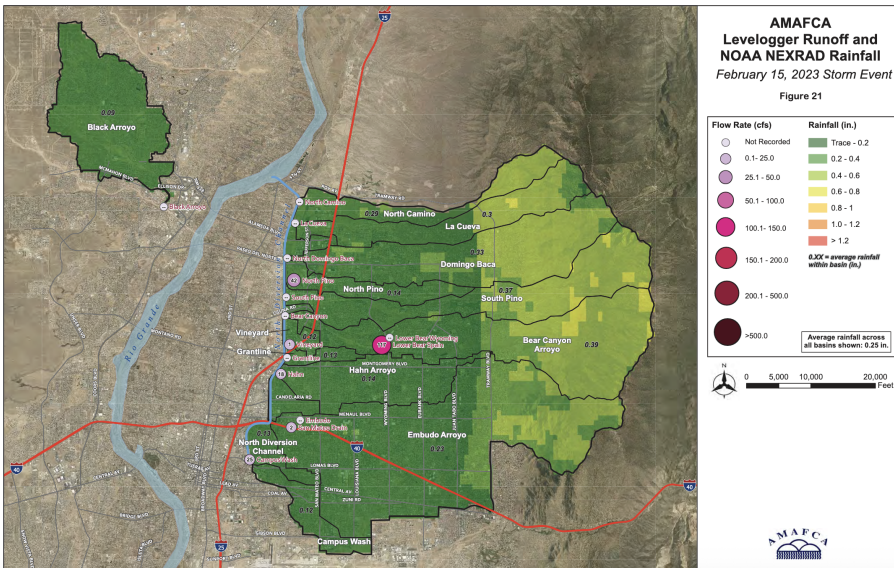
Storm runoff from streets, lawns, farms, and construction and industrial sites picks up pet waste, fertilizers, dirt, pesticides, oil, and grease etc. with minimal treatment before reaching Rio Grande.



WR Mgmt Discussion

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ABQ Metropolitan Arroyo Flood Control Authority



City of Albuquerque v. Browner



Floods

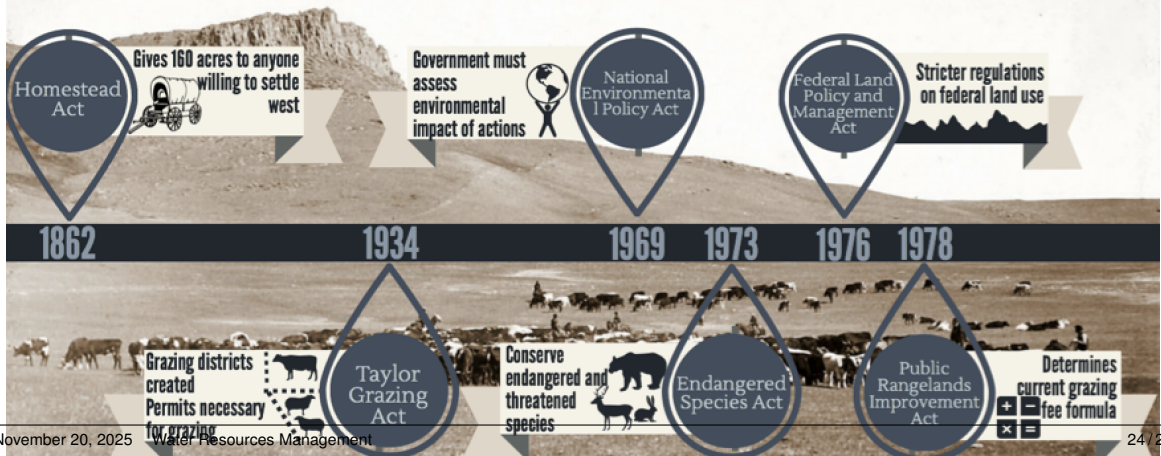


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Grazing Regulation

Regulations Effecting Grazing Lands



Middle Rio Grande Conservancy District

The MRGCD was created in 1923 to provide flood protection from the Rio Grande, drain swamp lands (from shallow water table), and provide irrigation water to farmlands. Partnered with Reclamation and Army Corps during Great Depression to accomplish larger infrastructure projects.

- El Vado (storage) and Cochiti (flood control) dams among others
- Dug and maintain 17 miles of drainage and irrigation canals to support agriculture
- Constructed riverside levees and jetties/checks for flood control
- Channelized the river

Habitat Restoration

Goals:

- Reduce the cover of non-native vegetation
- Increase the structural diversity of floodplain vegetation communities
- Improve river-floodplain connectivity
- Protect, extend, and enhance areas of potential habitat for listed species
- Reduce the risk of high severity fires in the bosque



Systematic Stream Gaging invented in New Mexico



Rio Grande at Embudo

